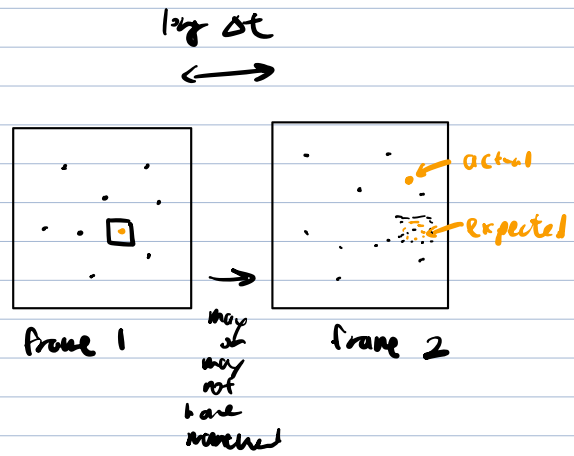

Literature:

- **Detection:**
  - Detection when your predicted state estimate and actual measurement (residual)
  - **NIS** (Normalized Innov. Squared) > 3 sigma
    "how off are you"

Detection Triggers:

- Generate a bank of hypotheses
- Traditionally, ad hoc
  - estimate of their ΔV budget
  - estimate of when maneuvered

- Overall traditional way: MHT (Multiple hypothesis tracking)

Project Idea (high level):

Optimally allocate compute to viable hypotheses

- Can we use sampling based methods to generate hypotheses (e.g. MCTS or others)
- Can we combine with physics-based constraints (eg. reachability) to prune as you go.

Two player Martian relief game:

Agent 1: You

Agent 2: maneuvering sat. ⇒

Agent 2: wants to decorrelate your filter

Agent 1: want to converge your filter

A lot of questions: Formalize the dynamics

- J2 dynamics nominally
- Impulse maneuvers only

- **High level:**

⇒ How to optimally generate & prune impulse maneuver hypotheses s.t. we accurately reconstruct a maneuver and task our sensors to maintain custody of tracked objects.